Jayprakash S. Nair

M.Tech. in Artificial Intelligence & Data Science
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Summary

Researcher in Multi-Agent Multi-Robot Systems. My current research focuses on **Autonomous Systems**, **Federated Reinforcement Learning**, **Swarm Learning**, **Evolutionary Algorithms**, and **Genetic Programming**. I have to my credit, three papers (two of them as the first author) on decentralized learning strategies in **Multi-Robot** environments using **mobile agents**. I aim to contribute to innovative **Lifelong Bio-Inspired Learning** solutions in the realm of robotics.

Education (in english medium throughout)

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (AI & DS)	Indian Institute of Technology (IIT) Patna	9.09 (Current)	2024-June 2026
B.Tech. (CSE)	APJ Abdul Kalam Technological University, Kerala, India	7.05	2018-2022
Senior Secondary	Central Board of Secondary Education (CBSE), India (Physics, Comp.Sc., Maths., Chem. & English)	76.2%	2018

Research Interests

- Multi-Agent Systems, Multi-Robot Learning, Federated Reinforcement Learning, Swarm Intelligence, Autonomous UAVs, Evolutionary Computation, Bio-inspired AI, Lifelong Learning

Publications

** denotes equal contribution*

C1. Jayprakash S. Nair, Jimson Mathew, and S. B. Nair, "A Reinforcement Learning-Inspired Latent Yield-based Adaptive Algorithm Switching Mechanism", *EvoStar 2026 (CORE Ranked Conference)*, Toulouse, France, 8-12th April 2026. **To Appear in:**. *Springer EvoApplications 2026 Part 1*

C2. Jayprakash S. Nair, Divya D. Kulkarni, Ajitem Joshi, and Sruthy Suresh, "On Decentralizing Federated Reinforcement Learning in Multi-Robot Scenarios", *7th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)*, 2022. **Citations: 18**.
DOI: *10.1109/SEEDA-CECNSM57760.2022.9932985*.

C3. Gayathri Rangu*, Divya D. Kulkarni*, **Jayprakash S. Nair***, and S. B. Nair, '"A Hybrid Federated Reinforcement Learning Approach for Networked Robots", *Recent Advances in Electrical and Electronic Engineering*, Springer, 2023.
Citations: 3.
DOI: *10.1007/1978-981-99-4713-3_47*.

Professional and Research Experience

- **Research Intern** *Sep. 2025 - Present*
  Indian Institute of Technology (IIT) Ropar Ropar, India
  – Developing visual navigation for drones

- **Software Engineer** *July 2022 - Aug. 2024*
  People10 Technologies Inc. Bengaluru, India
  – Developed e-commerce features for Allegiant Air (GraphQL, pricing, bundles, bookings)
  – Contributed to UI enhancement, GraphQL mutation development, shopping cart pricing updates, and QA process automation for the Allegiant Air project under the Navitaire Initiative
  – **Tools & technologies used**: PHP, Symfony, GraphQL and JavaScript

- **Research Intern** *Feb. 2019 - Dec. 2019*
  Indian Institute of Technology (IIT) Guwahati Guwahati, India
  – Learnt to control robots within Webots using Python
  – Program mobile agents using *Tartarus*, a multi-mobile agent platform

Projects

- **Open-Source Multi-Mobile-Agent Platform (Tartarus in Python)** *2025 - In Progress*
  Indian Institute of Technology Guwahati Website
  - Developing a Python-based library to bring the Tartarus multi-mobile agent platform to Python developers for distributed multi-agent systems
  - Enables distributed control, agent mobility, and coordination across heterogeneous networked nodes
  - Supports multi-threaded execution and cross-platform interoperability for scalable multi-agent environments
  - **Tools & technologies used**: Python (Janus), Prolog, Multi-threading, Distributed Systems

- **Visual Navigation for drones** *Sep. 2025 - Nov. 2025*
  Indian Institute of Technology Ropar
  - Worked on developing a Visual Navigation System for a drone to enable autonomous operation in GPS-denied environments
  - Utilized preloaded satellite imagery along with real-time visual data from the drone
  - Performed image comparison between satellite and drone imagery to guide navigation
  - **Tools & technologies used**: OpenCV, ViT models, and SIFT models

- **Population Based Training for a Multi-Robotic scenario** *Jul. 2025 - In Progress*
  Indian Institute of Technology Patna
  - Proposes a latent yield–based method for stable and adaptive online algorithm selection
  - Uses island models to enable parallel exploration and performance exchange among algorithms
  - Demonstrates the approach's effectiveness on sorting and robotic obstacle-avoidance tasks, showing improved adaptability and stability
  - **Tools & technologies used**: Python and Webots

- **On Decentralizing Federated Reinforcement Learning in Multi-Robot Scenarios** *Dec. 2021 - Jan. 2022*
  Bachelor's Project
  - Implemented Decentralized Federated RL (Q- & SARSA) for robots running on multiple instantiations of Webots running on different PCs connected via the LAN, using mobile agents
  - **Tools & technologies used**: *Tartarus*, Python and Webots

- **RL for e-puck within Webots** *Feb. 2019 - Dec. 2019*
  IIT Guwahati Github
  - Developed and evaluated RL (Q- and SARSA) robot controllers in Webots
  - **Tools & technologies used**: Python and Webots

Key Courses Taken

- Deep Learning, Internet of Things, Natural Language Processing, Drone Building Course (Audit), Reinforcement Learning, Generative AI, Probability and Statistics, Linear Algebra, Data Structures, Algorithms, DBMS, Machine Learning, Artificial Intelligence, Web Services, Operating Systems

Technical Skills

- **Programming:** C/C++, Python(Advanced), R, JavaScript, SQL
- **Tools & OS:** Git, Jupyter Notebook, Google Colab, Webots, LaTex, Linux, Windows
- **Libraries/Frameworks:** Pandas, Numpy, Scikit-Learn, PyTorch (Intermediate–Advanced), Symfony, OpenCV, ViT Models, Matplotlib, Seaborn
- **Web Skills:** HTML/CSS/JS, ReactJS, PHP, GraphQL

Languages known

- English (Proficient in both written and spoken English), Hindi